Approach–avoidance tendencies and physical activity level: A systematic review and meta-analysis

Jabouille F, Boisgontier MP

29 of October, 2026

Contents

R Markdown file set-up **2**

Data file glimpse (not included in PDF) **2**

Meta-analysis method **2**

Primary analysis using metafor to compare models . . . 3

Explore heterogeneity . . . 4

Comparing models . . . 4

Primary analysis using metacor . . . 5

Secondary analysis based on Spearman's rho values **7**

Secondary meta-analysis of rho values with metacor . . . 7

Meta-analysis: subgroup analyses **8**

Subgroup analysis by physical activity objectivity measurement . . . 8

Subgroup analysis by Stimuli Type . . . 11

Subgroup analysis by testing context . . . 14

Subgroup analysis by approach-avoidance measures . . . 17

Meta-analysis: meta-regression (19 clusters) **21**

Meta-regression by age . . . 21

Meta-regression by proportion of women . . . 22

Sensitivity analysis: meta-regression by quality score **24**

Publication bias analysis **27**
Small-study effects 27
Funnel plot 27
Egger's test 30
Limit meta-analysis 32
Pcurve analysis 32

References **34**

R Markdown file set-up

Packages required: 1) dmetar 2) tidyverse 3) meta 4) metafor 5) metasens 6) esc 7) dplyr 8) stringr

Data file glimpse (not included in PDF)

Meta-analysis method

In the main meta-analysis, we pooled Pearson product–moment correlations from eligible studies to examine the relationship between approach–avoidance tendencies and physical activity. When exact p-values were not reported in an article, but the sample size (n) and Pearson's correlation coefficient were available, the exact p-value was computed using an ad-hoc R code. Correlations were pooled using the generic inverse-variance method via the metacor function in the R meta package (Schwarzer et al., 2023a). This function automatically performs the necessary Fisher's z-transformation on the original, untransformed correlations prior to pooling and reconverts the pooled estimate back to the original correlation metric for ease of interpretation. When multiple correlations originated from the same study (e.g., Fessler, 2024), we accounted for within-study dependence using two complementary approaches. First, we employed a three-level model implemented in the metafor package (Viechtbauer, 2010, 2023) with random effects specified as ~ 1 | Author/Cluster. Second, we conducted a parallel meta-analysis using the meta package (metacor, cluster = Author) with Knapp–Hartung adjustments to obtain cluster-robust confidence intervals. We anticipated considerable between-study heterogeneity and therefore used a random-effects model to pool the correlations. The restricted maximum likelihood (REML) estimator (Viechtbauer et al., 2005) was used to estimate the heterogeneity variance (^{2}). To further quantify heterogeneity, we reported the I^2 statistic, which represents the percentage of variability in observed correlations not attributable to sampling error (Higgins et al., 2002). The I^2 values were interpreted as follows: 0–40%, might not be important; 30–60%, may represent moderate heterogeneity; 50–90%, substantial heterogeneity; and 75–100%, considerable heterogeneity. To reduce the risk of false positives, we applied the Knapp–Hartung adjustment (Knapp et al., 2003) when calculating the confidence interval around the pooled association. We also report the prediction interval, which provides a range within which future studies' effects are expected to fall given the current evidence. The pooled correlation was interpreted according to Cohen's (1988) conventions: r 0.10 (small), r 0.30 (moderate), and r 0.50 (large).

Egger's regression test of funnel plot asymmetry (Egger et al., 1997) and a p-curve analysis (Simonsohn et al., 2014) were conducted to assess potential publication bias in our meta-analysis. The Rücker's limit meta-analysis method (Schwarzer et al., 2023b), which explicitly includes the heterogeneity variance in the model, was used to compute bias-corrected estimate of the true effect size.

A secondary meta-analysis was conducted using the same approach, but based on Spearman's rho values, to further test the relationship between approach-avoidance tendencies and physical activity.

Subgroup meta-analyses were conducted to examine differences in correlations between studies using different testing contexts (in-person vs. online), types of approach-avoidance tasks (Manikin task vs. modified Manikin task vs. Visual Approach-Avoidance by the Self Task (VAAST) vs. Implicit Association Test-Recoding Free (IAT-RF) vs. joystick-based approach-avoidance task), stimulus type (pictograms vs. words vs. images), and types of physical activity measures (self-reported vs. device-based).

Meta-regressions were conducted to examine if the average age of participants, the proportion of women, or the quality score of a study predicted the reported correlation between approach-avoidance tendencies and physical activity.

All analyses were conducted in RStudio integrated development environment (IDE) (2025.09.0 Build 387, "Cucumberleaf Sunflower" release) running R 4.4.2 (R Core Team, 2024), using the 'meta' (Schwarzer, 2023), 'metasens' (Schwarzer et al., 2023), and 'metafor' (Viechtbauer, 2010; Viechtbauer, 2023) R packages.

Primary analysis using metafor to compare models

Cluster represents the independent groups, whereas Effect_ID represents the effect sizes nested within these groups.

```
mv.cor <- rma.mv(
  yi     = AA_r$z,
  V      = AA_r$var.z,
  data   = AA_r,
  slab   = AA_r$Cluster,
  random = ~ 1 | Cluster/Effect_ID,
  test   = "t",
  method = "REML"
)
summary(mv.cor)
```

```
##
## Multivariate Meta-Analysis Model (k = 22; method: REML)
##
##   logLik  Deviance       AIC       BIC      AICc
##   6.8110  -13.6220   -7.6220   -4.4884   -6.2102
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed             factor
## sigma^2.1  0.0101  0.1006     19     no            Cluster
## sigma^2.2  0.0000  0.0000     22     no  Cluster/Effect_ID
##
## Test for Heterogeneity:
## Q(df = 21) = 41.8744, p-val = 0.0044
##
## Model Results:
##
## estimate      se    tval  df    pval   ci.lb   ci.ub
##   0.1090  0.0336  3.2486  21  0.0038  0.0392  0.1788  **
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
est_z <- as.numeric(coef(mv.cor))
ci_z  <- as.data.frame(confint(mv.cor))[, c("ci.lb", "ci.ub")][1, ]

c(est = round(metafor::transf.ztor(est_z), 2),
  lcl = round(metafor::transf.ztor(ci_z[1]), 2),
  ucl = round(metafor::transf.ztor(ci_z[2]), 2))
```

```
##       est lcl.ci.lb ucl.ci.ub
##      0.11      0.00      0.04
```

Explore heterogeneity

```r
i2 <- var.comp(mv.cor)
summary(i2)
```

```
##         % of total variance    I2
## Level 1        5.177947e+01   ---
## Level 2        3.400609e-07     0
## Level 3        4.822053e+01 48.22
## Total I2: 48.22%
```

The multilevel variance decomposition indicated that sampling error variance (Level 1) accounted for 51.78% of the total variance, while within-study heterogeneity (Level 2) was negligible ($I^2 = 0\%$). The share of variance originated from between-study heterogeneity (Level 3) represented 48.22% of the total variance. Overall, these results suggest that heterogeneity was primarily driven by differences between studies rather than by variability within studies.

Comparing models

Reduced model in which the level 3 variance (between-study heterogeneity) is set to 0, which assumes all effect sizes are independent.

```r
l3.removed <- rma.mv(yi = z,
                     V = var.z,
                     slab = Cluster,
                     data = AA_r,
                     random = ~ 1 | Cluster/Effect_ID,
                     test = "t",
                     method = "REML",
                     sigma2 =  c(0, NA))

summary(l3.removed)
```

```
##
## Multivariate Meta-Analysis Model (k = 22; method: REML)
##
##   logLik  Deviance       AIC       BIC      AICc
##   6.7510  -13.5020   -9.5020   -7.4130   -8.8354
##
```

```
## Variance Components:
##
##             estim    sqrt  nlvls  fixed             factor
## sigma^2.1  0.0000  0.0000     19    yes            Cluster
## sigma^2.2  0.0098  0.0992     22     no  Cluster/Effect_ID
##
## Test for Heterogeneity:
## Q(df = 21) = 41.8744, p-val = 0.0044
##
## Model Results:
##
## estimate      se    tval  df    pval   ci.lb   ci.ub
##   0.1066  0.0329  3.2448  21  0.0039  0.0383  0.1750  **
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Comparing full and reduced models.

```
anova(mv.cor, l3.removed)
```

```
##
##         df     AIC     BIC    AICc logLik    LRT   pval      QE
## Full     3 -7.6220 -4.4884 -6.2102 6.8110               41.8744
## Reduced  2 -9.5020 -7.4130 -8.8354 6.7510 0.1199 0.7291 41.8744
```

The comparison between the 2-level and 3-level models did not provide evidence to retain the more complex structure. The 2-level model showed slightly better fit indices (AIC = −9.50 vs. −7.62; BIC = −7.41 vs. −4.49), and the likelihood ratio test indicated that the improvement offered by the 3-level model was not statistically significant (^{2} = 0.12, p = .73). However, given that several studies contributed multiple effect sizes, the 3-level model was nevertheless retained on methodological grounds, as it more appropriately accounts for the dependence among estimates and avoids violating the assumption of independence.

Primary analysis using metacor

```
m.cor <- metacor(
  cor = Cor, n = n,
  studlab = paste(Cluster),
  data = AA_r,
  random = TRUE, common = FALSE,
  method.tau = "REML",
  hakn = TRUE,
  cluster = Cluster,
  prediction = TRUE,
  title = "Primary meta-analysis using metacor"
)
```

```
## Warning: Use argument 'method.random.ci' instead of 'hakn' (deprecated).
```

```r
summary(m.cor)
```

```
## Review:     Primary meta-analysis using metacor
##
##                       COR            95%-CI %W(random)          cluster
## Cheval, 2014       0.0600 [-0.1478; 0.2627]        5.2     Cheval, 2014
## Cheval, 2015       0.2700 [ 0.0746; 0.4455]        5.4     Cheval, 2015
## Cheval, 2016      -0.0600 [-0.2413; 0.1253]        5.9     Cheval, 2016
## Daou, 2017        -0.0230 [-0.2045; 0.1600]        5.9       Daou, 2017
## Cheval, 2018       0.2600 [-0.1177; 0.5720]        2.3     Cheval, 2018
## Oliver, 2018       0.0300 [-0.1645; 0.2222]        5.6     Oliver, 2018
## Hannan, 2019       0.2500 [ 0.0603; 0.4223]        5.6     Hannan, 2019
## Zenko, 2019        0.3230 [ 0.1266; 0.4951]        5.3      Zenko, 2019
## Berry, 2020        0.1870 [ 0.0270; 0.3376]        6.6      Berry, 2020
## Preis, 2021       -0.0230 [-0.3489; 0.3079]        2.2      Preis, 2021
## Preis, 2021       -0.0820 [-0.3998; 0.2534]        2.2      Preis, 2021
## Locke, 2021        0.0300 [-0.1416; 0.1998]        6.3      Locke, 2021
## Wang, 2023         0.2700 [-0.0933; 0.5698]        2.5       Wang, 2023
## Farajzadeh, 2025 -0.0400 [-0.1422; 0.0631]        8.7 Farajzadeh, 2025
## Fessler, 2024a     0.0400 [-0.2672; 0.3398]        2.4   Fessler, 2024a
## Fessler, 2024b     0.4200 [-0.1704; 0.7885]        0.9   Fessler, 2024b
## Fessler, 2024a     0.1200 [-0.1799; 0.3995]        2.6   Fessler, 2024a
## Fessler, 2024b    -0.2400 [-0.6575; 0.2902]        1.2   Fessler, 2024b
## Terashima, 2025    0.1500 [ 0.0199; 0.2751]        7.7  Terashima, 2025
## Quossi, 2025       0.0280 [-0.1108; 0.1658]        7.4    Quossi, 2025
## Wang, 2025         0.5850 [ 0.3077; 0.7707]        2.7       Wang, 2025
## Woodman, 2025      0.0700 [-0.1391; 0.2731]        5.2    Woodman, 2025
##
## Number of studies: n = 19
## Number of estimates: k = 22
## Number of observations: o = 2161
##
##                          COR            95%-CI    t p-value
## Random effects model 0.1086 [ 0.0392; 0.1769] 3.25  0.0038
## Prediction interval         [-0.1111; 0.3182]
##
## Quantifying heterogeneity (with 95%-CIs):
##  tau^2.1 = 0.0101 [0.0000; 0.0363]; tau.1 = 0.1006 [0.0000; 0.1906] (between cluster)
##  tau^2.2 < 0.0001 [0.0000; 0.0344]; tau.2 < 0.0001 [0.0000; 0.1855] (within cluster)
##  I^2 = 49.9% [18.0%; 69.3%]; H = 1.41 [1.10; 1.81]
##
## Test of heterogeneity:
##      Q d.f. p-value
##  41.87   21  0.0044
##
## Details of meta-analysis methods:
## - Inverse variance method (three-level model)
## - Restricted maximum-likelihood estimator for tau^2
## - Profile-Likelihood method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Random effects confidence interval based on t-distribution (df = 21)
## - Prediction interval based on t-distribution (df = 21)
## - Fisher's z transformation of correlations
```

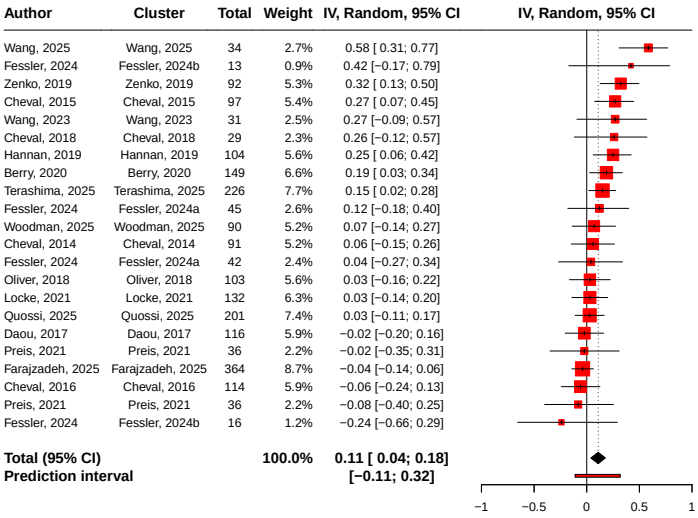
Our meta-analysis of 18 studies (n = 2,070) using a three-level random-effects model showed a statistically significant small positive correlation between approach–avoidance tendencies and physical activity (r = 0.11; 95% CI [0.04, 0.18]; p = .004).

Heterogeneity analyses showed that most of the variability came from differences between studies (tau^2_between = 0.0106; tau^1 = 0.1028), while variation within studies was almost nonexistent (tau^2_within < 0.0001). In other words, studies tended to differ from one another rather than producing inconsistent results internally. Overall heterogeneity was moderate to substantial (I^2 = 49.9%, 95% CI [18.0%, 69.3%]; Q(21) = 41.87, p = .0044), suggesting that factors like measurement methods or participant characteristics likely influenced the size of the observed effects.

The prediction interval (ranging from r = −0.11 to 0.32) further indicates that although the average effect is positive, future research could still find weaker, null, or even negative associations depending on the context. Taken together, these results point to a relationship that is statistically reliable but modest in size, emphasizing the need to consider heterogeneity when interpreting and generalizing these findings.

```
## Warning in forest.meta(m.cor, layout = "Revman5", sortvar = -TE, prediction =
## TRUE, : Information on between-study variance printed for RevMan5 layout
```

```
knitr::include_graphics("figures/figure-latex/Forest plot AA.pdf")
```



Secondary analysis based on Spearman's rho values

Secondary meta-analysis of rho values with metacor

```
m.cor.rho <- metacor(cor = Cor,
                     n = n,
                     data = AA_rho,
                     studlab = Cluster,
                     fixed = FALSE,
                     random = TRUE,
                     method.tau = "REML",
                     hakn = TRUE,
                     prediction = TRUE)
```

```
## Warning: Use argument 'common' instead of 'fixed' (deprecated).

## Warning: Use argument 'method.random.ci' instead of 'hakn' (deprecated).
```

```
summary(m.cor.rho)
```

```
##                            COR            95%-CI %W(random)
## Moffitt, 2020 Study 1 -0.1200 [-0.2583; 0.0231]       68.6
## Moffitt, 2020 Study 2  0.0100 [-0.1999; 0.2190]       31.4
##
## Number of studies: k = 2
## Number of observations: o = 277
##
##                          COR            95%-CI     t p-value
## Random effects model -0.0795 [-0.6908; 0.5981] -1.31  0.4141
## Prediction interval          [-0.6918; 0.5994]
##
## Quantifying heterogeneity:
##  tau^2 = 0; tau = 0; I^2 = 0.0%; H = 1.00
##
## Test of heterogeneity:
##     Q d.f. p-value
##  0.99    1  0.3186
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 1)
## - Prediction interval based on t-distribution (df = 1)
## - Fisher's z transformation of correlations
```

The secondary analysis using Spearman's rho (k = 2, n = 277) showed a very small and non-significant negative correlation (r = –0.08, 95% CI [–0.69, 0.60], p = 0.41). No heterogeneity was detected (^{2} = 0.00; I^2 = 0%), which is expected with only two studies. The prediction interval (–0.69 to 0.60) indicates that future research may still observe either negative, null, or positive effects, highlighting the need for more data before firm conclusions can be drawn.

Meta-analysis: subgroup analyses

Subgroup analysis by physical activity objectivity measurement

```
AA_r$PA_objectivity <- as.factor(AA_r$PA_objectivity)

PAobj_subg <- update(m.cor,
            subgroup = PA_objectivity,
            tau.common = TRUE)
PAobj_subg
```

```
## Review:     Primary meta-analysis using metacor
##
## Number of studies: n = 19
## Number of estimates: k = 22
## Number of observations: o = 2161
##
##                          COR            95%-CI    t p-value
## Random effects model 0.1086 [ 0.0392; 0.1769] 3.25  0.0038
## Prediction interval         [-0.1111; 0.3182]
##
## Quantifying heterogeneity (with 95%-CIs):
##  tau^2.1 = 0.0101 [0.0000; 0.0363]; tau.1 = 0.1006 [0.0000; 0.1906] (between cluster)
##  tau^2.2 < 0.0001 [0.0000; 0.0344]; tau.2 < 0.0001 [0.0000; 0.1855] (within cluster)
##  I^2 = 49.9% [18.0%; 69.3%]; H = 1.41 [1.10; 1.81]
##
## Quantifying residual heterogeneity (with 95%-CIs):
##  tau^2 = 0.0109; tau = 0.1042; I^2 = 51.7%; H = 1.44
##
## Test of heterogeneity:
##      Q d.f. p-value
##  41.87   21  0.0044
##
## Results for subgroups (random effects model):
##                                          k    COR            95%-CI  tau^2
## PA_objectivity = Self-reported measure  17 0.1056 [ 0.0285; 0.1815] 0.0110
## PA_objectivity = Device-based measure    5 0.1280 [-0.0368; 0.2859] 0.0110
##                                           tau     Q   I^2
## PA_objectivity = Self-reported measure 0.1047 36.32 56.0%
## PA_objectivity = Device-based measure  0.1047  5.12 21.8%
##
## Test for subgroup differences (random effects model):
##                   Q d.f. p-value
## Between groups 0.07    1  0.7947
##
## Details of meta-analysis methods:
## - Inverse variance method (three-level model)
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Profile-Likelihood method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Random effects confidence interval based on t-distribution (df = 21)
## - Prediction interval based on t-distribution (df = 21)
## - Fisher's z transformation of correlations
```

A subgroup analysis was conducted to examine whether the association between approach-avoidance tendencies and physical activity differed in self-reported (k = 17) and device-based measures of physical activity (k = 5). Results showed no evidence of a moderation effect (Q(1) = 0.07, p = .795).

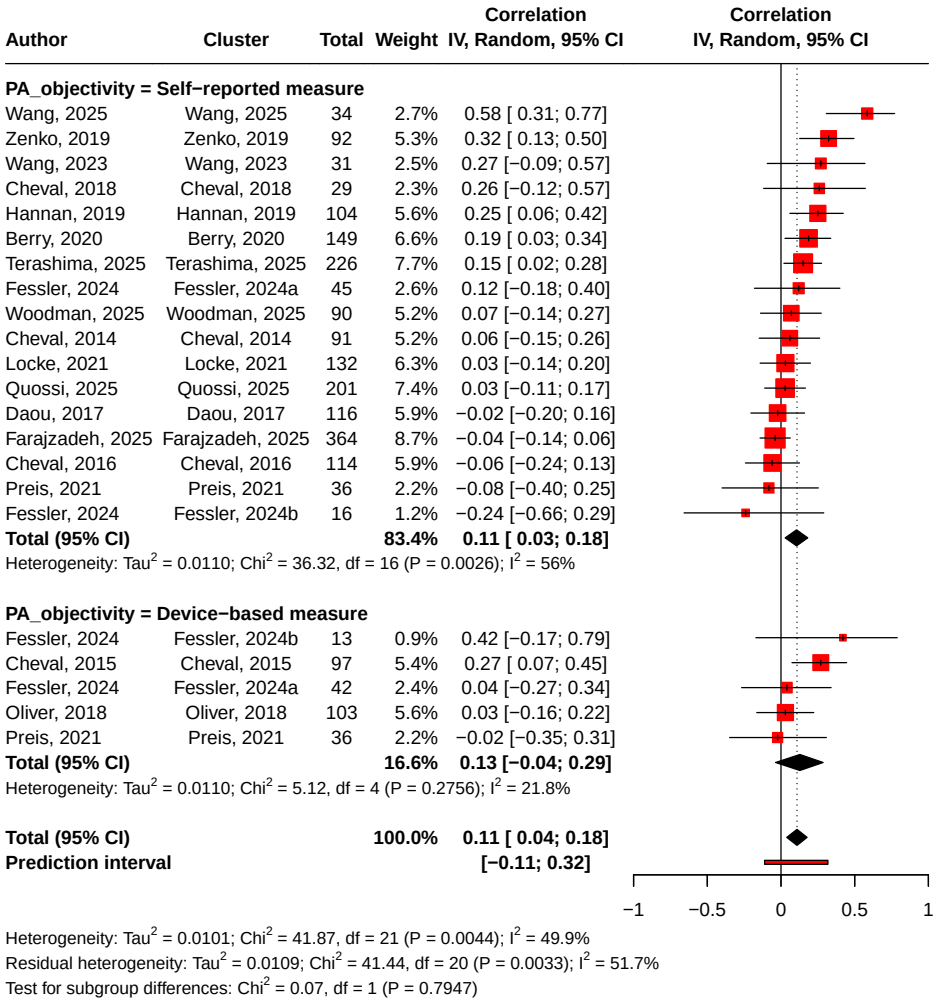
```
pdf("figures/figure-latex/PA_objectivity subgroup forest plot.pdf",
    width = 10, height = 11, onefile = FALSE)

forest(PAobj_subg,
            layout = "RevMan5",
            sortvar = -TE,
```

```
            leftcols = c("Author", "Cluster", "n", "w.random", "effect.ci"),
            leftlabs = c("Author", "Cluster", "Total", "Weight", "Correlation\nIV, Random, 95% CI"),
            common = FALSE,
            xlim = c(-1.0, 1.0),
            prediction = TRUE,
            fontsize = 11,
            fs.hetstat = 10,
            col.subgroup = 'black',
            addrows.below.overall = 2)

dev.off()
```

```
## pdf
##   2
```



Subgroup analysis by Stimuli Type

```
AA_r$Stimuli_type<- as.factor(AA_r$Stimuli_type)

stim_type <- update(m.cor,
            subgroup = Stimuli_type,
            tau.common = TRUE)
stim_type
```

```
## Review:     Primary meta-analysis using metacor
##
## Number of studies: n = 19
## Number of estimates: k = 22
## Number of observations: o = 2161
##
##                          COR            95%-CI    t p-value
## Random effects model 0.1086 [ 0.0392; 0.1769] 3.25  0.0038
## Prediction interval         [-0.1111; 0.3182]
##
## Quantifying heterogeneity (with 95%-CIs):
##  tau^2.1 = 0.0101 [0.0000; 0.0363]; tau.1 = 0.1006 [0.0000; 0.1906] (between cluster)
##  tau^2.2 < 0.0001 [0.0000; 0.0344]; tau.2 < 0.0001 [0.0000; 0.1855] (within cluster)
##  I^2 = 49.9% [18.0%; 69.3%]; H = 1.41 [1.10; 1.81]
##
## Quantifying residual heterogeneity (with 95%-CIs):
##  tau^2 = 0.0111; tau = 0.1052; I^2 = 51.0%; H = 1.43
##
## Test of heterogeneity:
##      Q d.f. p-value
##  41.87   21  0.0044
##
## Results for subgroups (random effects model):
##                              k    COR            95%-CI  tau^2    tau     Q
## Stimuli_type = Pictograms  15 0.0938 [ 0.0048; 0.1812] 0.0115 0.1073 29.38
## Stimuli_type = Words        2 0.1646 [-0.0542; 0.3683] 0.0115 0.1073  1.60
## Stimuli_type = Images       5 0.1303 [-0.0216; 0.2764] 0.0115 0.1073  7.79
##                              I^2
## Stimuli_type = Pictograms 52.3%
## Stimuli_type = Words      37.7%
## Stimuli_type = Images     48.7%
##
## Test for subgroup differences (random effects model):
##                   Q d.f. p-value
## Between groups 0.25    2  0.7786
##
## Details of meta-analysis methods:
## - Inverse variance method (three-level model)
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Profile-Likelihood method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Random effects confidence interval based on t-distribution (df = 21)
## - Prediction interval based on t-distribution (df = 21)
## - Fisher's z transformation of correlations
```

A subgroup meta-analysis was conducted to examine whether the association differed by stimulus type (pictograms vs. words vs. images). Results showed no evidence of a moderation effect (Q(2) = 0.25, p = .779).

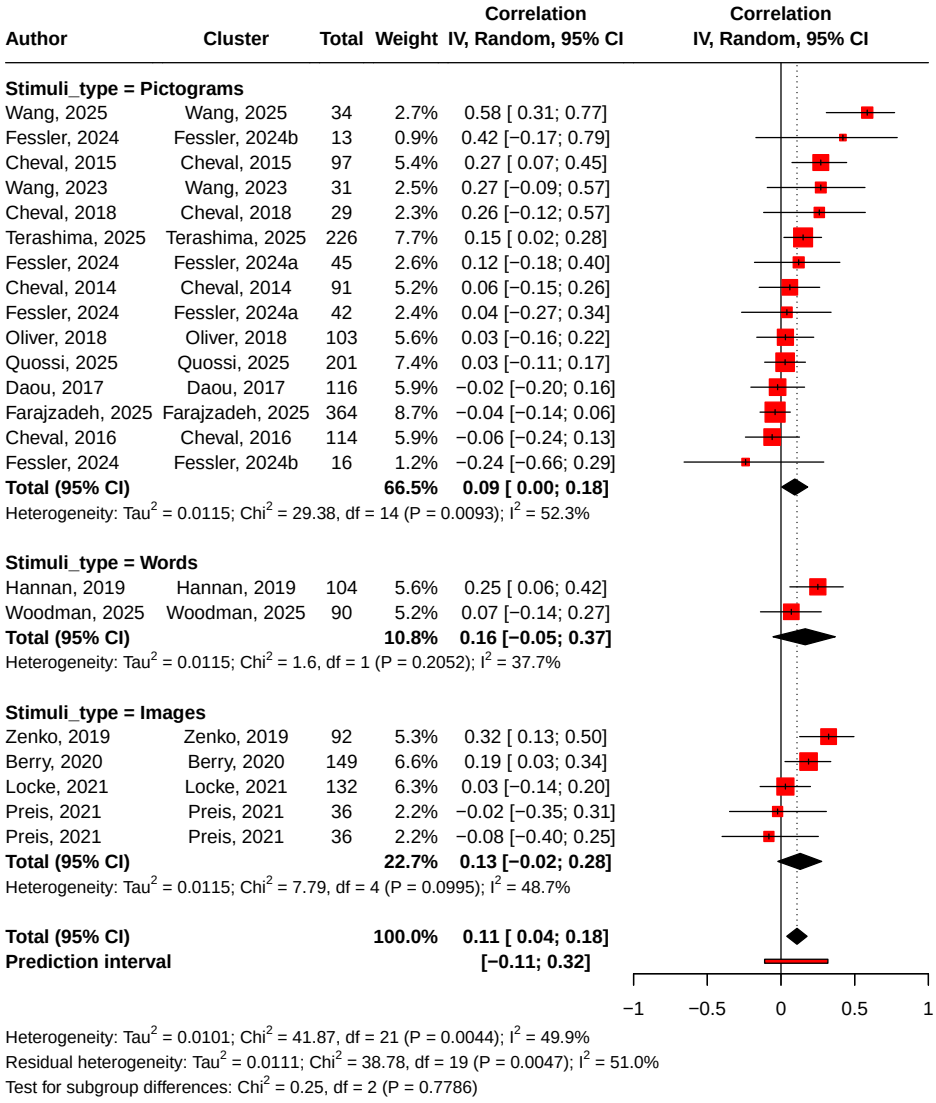
```
pdf("figures/figure-latex/stim_type subgroup forest plot.pdf",
    width = 10, height = 11, onefile = FALSE)

forest(stim_type,
```

```
            layout = "RevMan5",
            sortvar = -TE,
            leftcols = c("Author", "Cluster", "n", "w.random", "effect.ci"),
            leftlabs = c("Author", "Cluster", "Total", "Weight", "Correlation\nIV, Random, 95% CI"),
            common = FALSE,
            xlim = c(-1.0, 1.0),
            prediction = TRUE,
            fontsize = 11,
            fs.hetstat = 10,
            col.subgroup = 'black',
            addrows.below.overall = 2)

dev.off()
```

```
## pdf
##   2
```



Subgroup analysis by testing context

```
AA_r$Testing_context <- as.factor(AA_r$Testing_context)

Testing_context <- update(m.cor,
            subgroup = Testing_context,
            tau.common = TRUE)
```

```
Testing_context
```

```
## Review:     Primary meta-analysis using metacor
##
## Number of studies: n = 19
## Number of estimates: k = 22
## Number of observations: o = 2161
##
##                          COR            95%-CI    t p-value
## Random effects model 0.1086 [ 0.0392; 0.1769] 3.25  0.0038
## Prediction interval         [-0.1111; 0.3182]
##
## Quantifying heterogeneity (with 95%-CIs):
##  tau^2.1 = 0.0101 [0.0000; 0.0363]; tau.1 = 0.1006 [0.0000; 0.1906] (between cluster)
##  tau^2.2 < 0.0001 [0.0000; 0.0344]; tau.2 < 0.0001 [0.0000; 0.1855] (within cluster)
##  I^2 = 49.9% [18.0%; 69.3%]; H = 1.41 [1.10; 1.81]
##
## Quantifying residual heterogeneity (with 95%-CIs):
##  tau^2 = 0.0055; tau = 0.0742; I^2 = 39.1%; H = 1.28
##
## Test of heterogeneity:
##      Q d.f. p-value
##  41.87   21  0.0044
##
## Results for subgroups (random effects model):
##                               k    COR            95%-CI  tau^2    tau     Q
## Testing_context = In-person  16 0.1608 [ 0.0794; 0.2401] 0.0061 0.0782 26.76
## Testing_context = Online      6 0.0294 [-0.0646; 0.1228] 0.0061 0.0782  6.06
##                               I^2
## Testing_context = In-person 44.0%
## Testing_context = Online    17.5%
##
## Test for subgroup differences (random effects model):
##                   Q d.f. p-value
## Between groups 4.90    1  0.0387
##
## Details of meta-analysis methods:
## - Inverse variance method (three-level model)
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Profile-Likelihood method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Random effects confidence interval based on t-distribution (df = 21)
## - Prediction interval based on t-distribution (df = 21)
## - Fisher's z transformation of correlations
```

```
Testing_context$pval.random.w
```

```
##    In-person       Online
## 0.0005655457 0.5222428393
```

A subgroup meta-analysis was conducted to examine whether the association differed by testing context (in-person vs. online). Results showed a moderation by testing context (Q(1) = 4.90, p = .039), with a

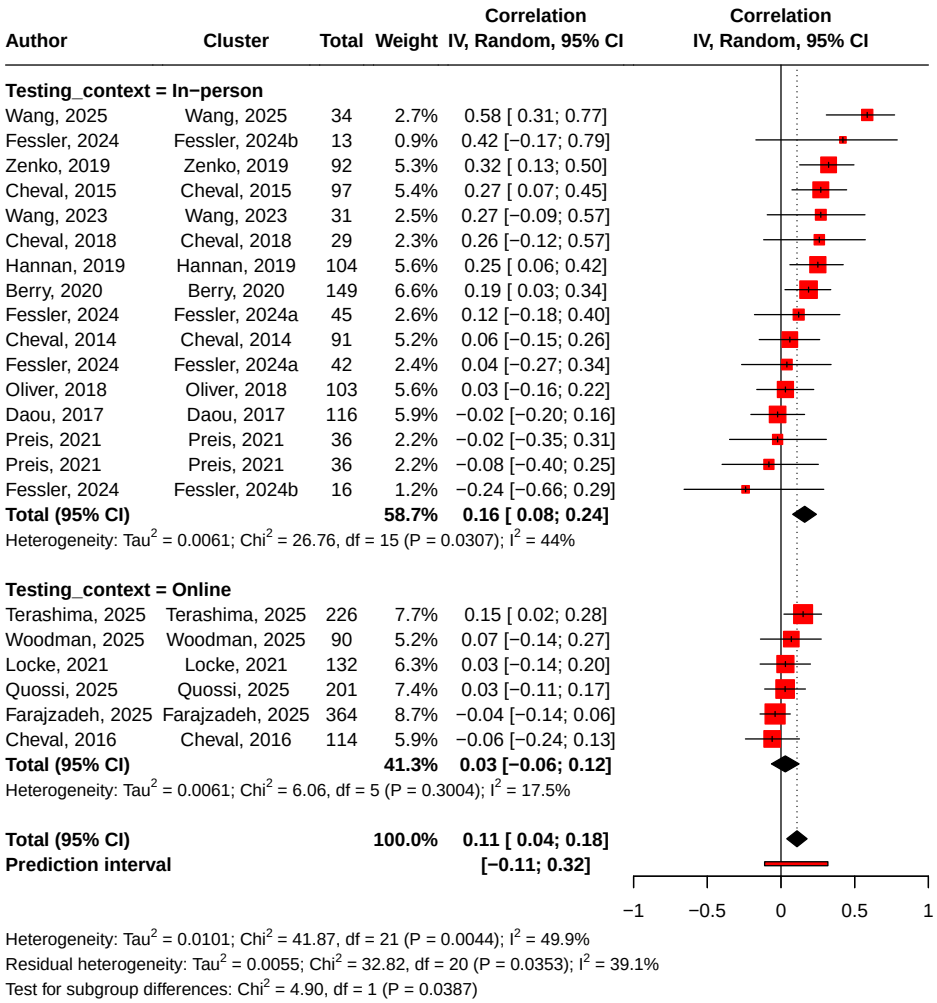
small positive association in in-person studies (r = .16, 95% CI: .08, .24, p = .0006), but not evidence of an association in online studies (r = .03, 95% CI: −.06, .12, p = .522)

```
pdf("figures/figure-latex/testing_context_subgroup_forest_plot.pdf",
    width = 10, height = 12, onefile = FALSE)

forest(Testing_context,
            layout = "RevMan5",
            sortvar = -TE,
            leftcols = c("Author", "Cluster", "n", "w.random", "effect.ci"),
            leftlabs = c("Author", "Cluster", "Total", "Weight", "Correlation\nIV, Random, 95% CI"),
            common = FALSE,
            xlim = c(-1.0, 1.0),
            prediction = TRUE,
            fontsize = 11,
            fs.hetstat = 10,
            col.subgroup = 'black',
            addrows.below.overall = 2)

dev.off()
```

```
## pdf
##   2
```



Subgroup analysis by approach-avoidance measures

```
AA_r$AAT_measure <- as.factor(AA_r$AAT_measure)
```

```r
AAT_subg <- update(m.cor,
            subgroup = AAT_measure,
            tau.common = TRUE)
AAT_subg
```

```
## Review:     Primary meta-analysis using metacor
##
## Number of studies: n = 19
## Number of estimates: k = 22
## Number of observations: o = 2161
##
##                          COR            95%-CI    t p-value
## Random effects model 0.1086 [ 0.0392; 0.1769] 3.25  0.0038
## Prediction interval         [-0.1111; 0.3182]
##
## Quantifying heterogeneity (with 95%-CIs):
##  tau^2.1 = 0.0101 [0.0000; 0.0363]; tau.1 = 0.1006 [0.0000; 0.1906] (between cluster)
##  tau^2.2 < 0.0001 [0.0000; 0.0344]; tau.2 < 0.0001 [0.0000; 0.1855] (within cluster)
##  I^2 = 49.9% [18.0%; 69.3%]; H = 1.41 [1.10; 1.81]
##
## Quantifying residual heterogeneity (with 95%-CIs):
##  tau^2 = 0.0110; tau = 0.1049; I^2 = 52.0%; H = 1.44
##
## Test of heterogeneity:
##      Q d.f. p-value
##  41.87   21  0.0044
##
## Results for subgroups (random effects model):
##                                    k     COR            95%-CI  tau^2    tau
## AAT_measure = Manikin             12  0.1133 [ 0.0229; 0.2019] 0.0118 0.1088
## AAT_measure = Modified Manikin     1 -0.0600 [-0.3494; 0.2398] 0.0118 0.1088
## AAT_measure = IAT-RF               1  0.2500 [-0.0556; 0.5128] 0.0118 0.1088
## AAT_measure = Joystick-based AAT   3  0.1551 [-0.0788; 0.3728] 0.0118 0.1088
## AAT_measure = VAAST                5  0.0723 [-0.1354; 0.2739] 0.0118 0.1088
##                                      Q   I^2
## AAT_measure = Manikin            26.83 59.0%
## AAT_measure = Modified Manikin    0.00    --
## AAT_measure = IAT-RF              0.00    --
## AAT_measure = Joystick-based AAT  5.75 65.2%
## AAT_measure = VAAST               2.85  0.0%
##
## Test for subgroup differences (random effects model):
##                   Q d.f. p-value
## Between groups 0.67    4  0.6210
##
## Details of meta-analysis methods:
## - Inverse variance method (three-level model)
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Profile-Likelihood method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Random effects confidence interval based on t-distribution (df = 21)
## - Prediction interval based on t-distribution (df = 21)
```

```
## - Fisher's z transformation of correlations
```

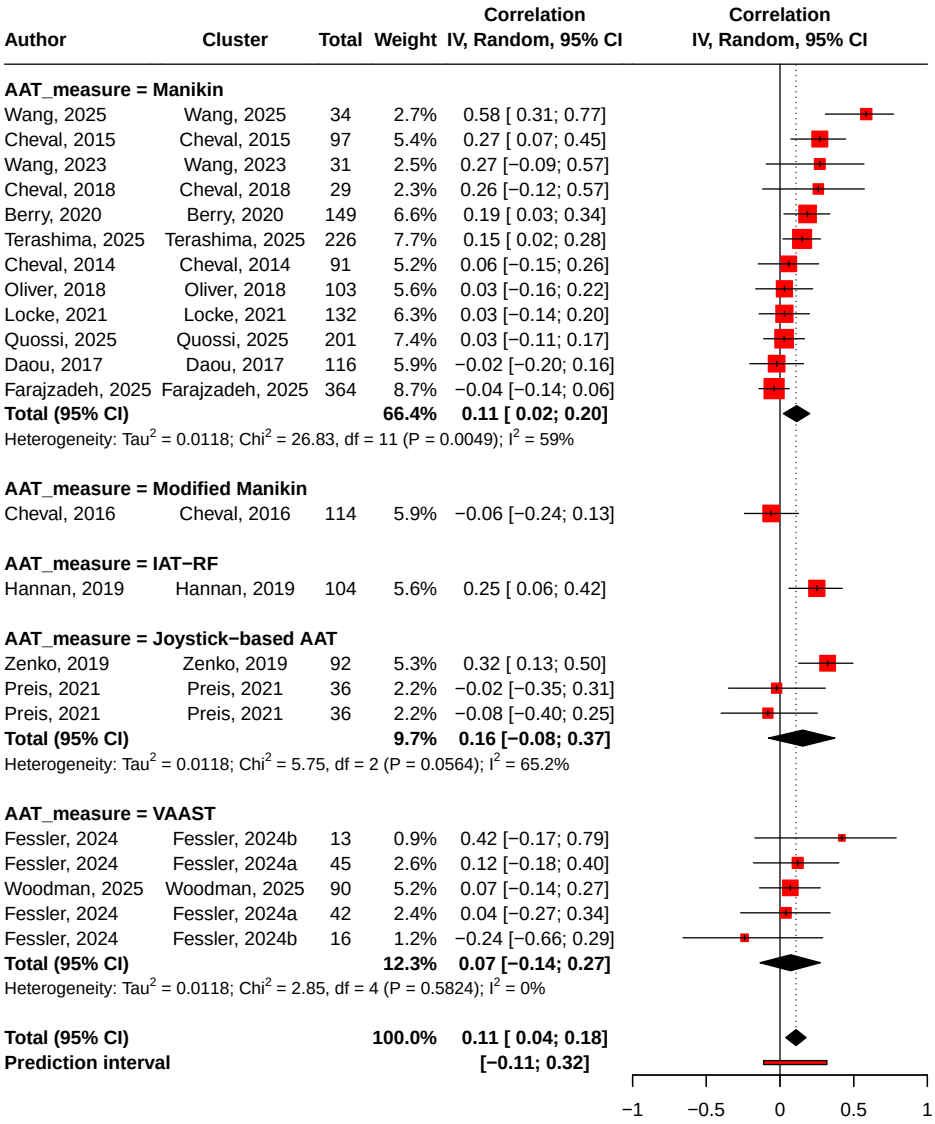
A subgroup analysis was conducted to compare the association between approach-avoidance tendencies and physical activity across different approach-avoidance tasks: the Manikin task (k = 12), the modified Manikin task (k = 1), the IAT-RF (k = 1), the joystick-based approach-avoidance tasks (k = 3), and the VAAST (k = 5). Results showed no evidence of a variation between tasks (Q(4) = 0.67, p = .621).

```
pdf("figures/figure-latex/AAT_subgroup_forest_plot.pdf",
    width = 10, height = 12, onefile = FALSE)

forest(AAT_subg,
            layout = "RevMan5",
            sortvar = -TE,
            leftcols = c("Author", "Cluster", "n", "w.random", "effect.ci"),
            leftlabs = c("Author", "Cluster", "Total", "Weight", "Correlation\nIV, Random, 95% CI"),
            common = FALSE,
            xlim = c(-1.0, 1.0),
            prediction = TRUE,
            fontsize = 11,
            fs.hetstat = 10,
            col.subgroup = 'black',
            addrows.below.overall = 2)

dev.off()
```

```
## pdf
##   2
```


Author
Cluster
Total
Weight
Correlation
IV, Random, 95% CI
Correlation
IV, Random, 95% CI
AAT_measure = Manikin
Wang, 2025 Wang, 2025 34 2.7% 0.58 [0.31; 0.77]
Cheval, 2015 Cheval, 2015 97 5.4% 0.27 [0.07; 0.45]
Wang, 2023 Wang, 2023 31 2.5% 0.27 [−0.09; 0.57]
Cheval, 2018 Cheval, 2018 29 2.3% 0.26 [−0.12; 0.57]
Berry, 2020 Berry, 2020 149 6.6% 0.19 [0.03; 0.34]
Terashima, 2025 Terashima, 2025 226 7.7% 0.15 [0.02; 0.28]
Cheval, 2014 Cheval, 2014 91 5.2% 0.06 [−0.15; 0.26]
Oliver, 2018 Oliver, 2018 103 5.6% 0.03 [−0.16; 0.22]
Locke, 2021 Locke, 2021 132 6.3% 0.03 [−0.14; 0.20]
Quossi, 2025 Quossi, 2025 201 7.4% 0.03 [−0.11; 0.17]
Daou, 2017 Daou, 2017 116 5.9% −0.02 [−0.20; 0.16]
Farajzadeh, 2025 Farajzadeh, 2025 364 8.7% −0.04 [−0.14; 0.06]
Total (95% CI) 66.4% 0.11 [0.02; 0.20]
Heterogeneity: Tau² = 0.0118; Chi² = 26.83, df = 11 (P = 0.0049); I² = 59%
AAT_measure = Modified Manikin
Cheval, 2016 Cheval, 2016 114 5.9% −0.06 [−0.24; 0.13]
AAT_measure = IAT−RF
Hannan, 2019 Hannan, 2019 104 5.6% 0.25 [0.06; 0.42]
AAT_measure = Joystick−based AAT
Zenko, 2019 Zenko, 2019 92 5.3% 0.32 [0.13; 0.50]
Preis, 2021 Preis, 2021 36 2.2% −0.02 [−0.35; 0.31]
Preis, 2021 Preis, 2021 36 2.2% −0.08 [−0.40; 0.25]
Total (95% CI) 9.7% 0.16 [−0.08; 0.37]
Heterogeneity: Tau² = 0.0118; Chi² = 5.75, df = 2 (P = 0.0564); I² = 65.2%
AAT_measure = VAAST
Fessler, 2024 Fessler, 2024b 13 0.9% 0.42 [−0.17; 0.79]
Fessler, 2024 Fessler, 2024a 45 2.6% 0.12 [−0.18; 0.40]
Woodman, 2025 Woodman, 2025 90 5.2% 0.07 [−0.14; 0.27]
Fessler, 2024 Fessler, 2024a 42 2.4% 0.04 [−0.27; 0.34]
Fessler, 2024 Fessler, 2024b 16 1.2% −0.24 [−0.66; 0.29]
Total (95% CI) 12.3% 0.07 [−0.14; 0.27]
Heterogeneity: Tau² = 0.0118; Chi² = 2.85, df = 4 (P = 0.5824); I² = 0%
Total (95% CI) 100.0% 0.11 [0.04; 0.18]
Prediction interval [−0.11; 0.32]
−1
−0.5
0
0.5
1
Heterogeneity: Tau² = 0.0101; Chi² = 41.87, df = 21 (P = 0.0044); I² = 49.9%
Residual heterogeneity: Tau² = 0.0110; Chi² = 35.44, df = 17 (P = 0.0054); I² = 52.0%
Test for subgroup differences: Chi² = 0.67, df = 4 (P = 0.6210)

Meta-analysis: meta-regression (19 clusters)

Meta-regression by age

```
m.cor.reg.age <- metareg(m.cor, ~Age)
m.cor.reg.age
```

```
##
## Multivariate Meta-Analysis Model (k = 22; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed    factor
## sigma^2.1  0.0104  0.1019     19     no        .id
## sigma^2.2  0.0000  0.0000     22     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 20) = 38.2833, p-val = 0.0082
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 20) = 0.7318, p-val = 0.4024
##
## Model Results:
##
##          estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt    0.1784  0.0875   2.0376  20  0.0550  -0.0042  0.3609  .
## Age       -0.0023  0.0027  -0.8555  20  0.4024  -0.0079  0.0033
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

A meta-regression was conducted to examine whether participants' mean age moderated the relationship between approach–avoidance tendencies and physical activity. Results indicated that age did not significantly predict the effect sizes, b = −0.002, 95% CI [−0.008, 0.003], t(20) = −0.86, p = .402.

```
pdf("figures/figure-latex/Bubble plot for meta-regression by age.pdf",
    width = 6, height = 6, onefile = FALSE)

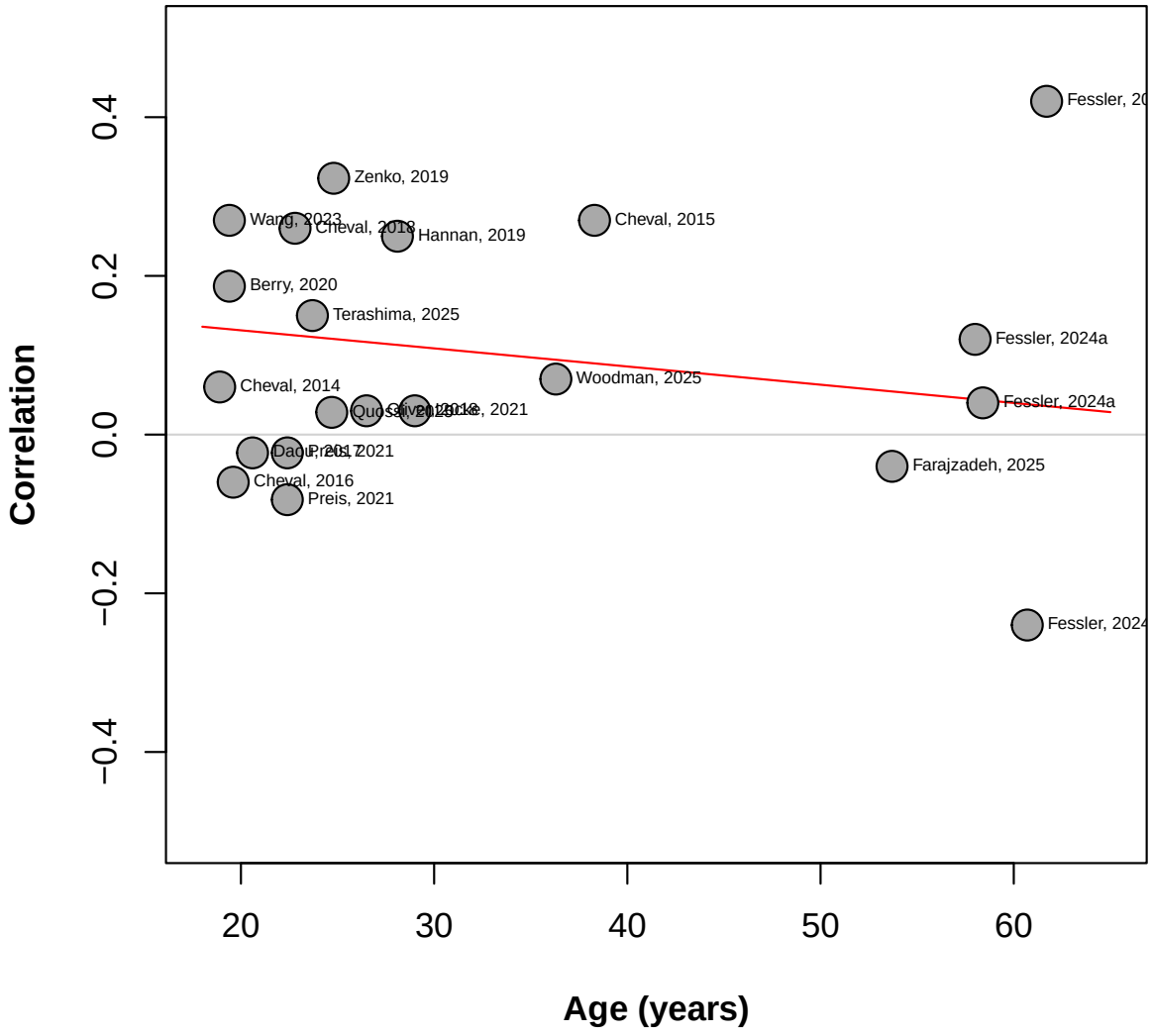
Age.bubble <- bubble(m.cor.reg.age,
                     xlim = c(18, 65),
                     ylim = c(-0.5, 0.5),
                     xlab = 'Age (years)',
                     font.lab = 2,
                     studlab = TRUE,
                     cex = 2,
                     cex.studlab = 0.5,
                     pos.studlab = 4,
                     offset = 0.5,
                     col.line = 'red')

Age.bubble
```

```
## NULL
```

```
dev.off()
```

```
## pdf
##   2
```



Meta-regression by proportion of women

```
m.cor.reg.women <- metareg(m.cor, ~Prop_women)
m.cor.reg.women
```

```
##
## Multivariate Meta-Analysis Model (k = 22; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed    factor
## sigma^2.1  0.0112  0.1059     19     no        .id
## sigma^2.2  0.0000  0.0000     22     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 20) = 41.7554, p-val = 0.0030
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 20) = 0.1983, p-val = 0.6609
##
## Model Results:
##
##             estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt       0.1918  0.1869   1.0263  20  0.3170  -0.1981  0.5818
## Prop_women   -0.1273  0.2859  -0.4453  20  0.6609  -0.7236  0.4690
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The proportion of women did not statistically influence the correlation values of the meta-analysis studies (b = −0.13, 95% CI [−0.72, 0.47], t(20) = −0.45, p = 0.661).

```
pdf("figures/figure-latex/Bubble plot for meta-regression by proportion of women.pdf",
    width = 6, height = 6, onefile = FALSE)

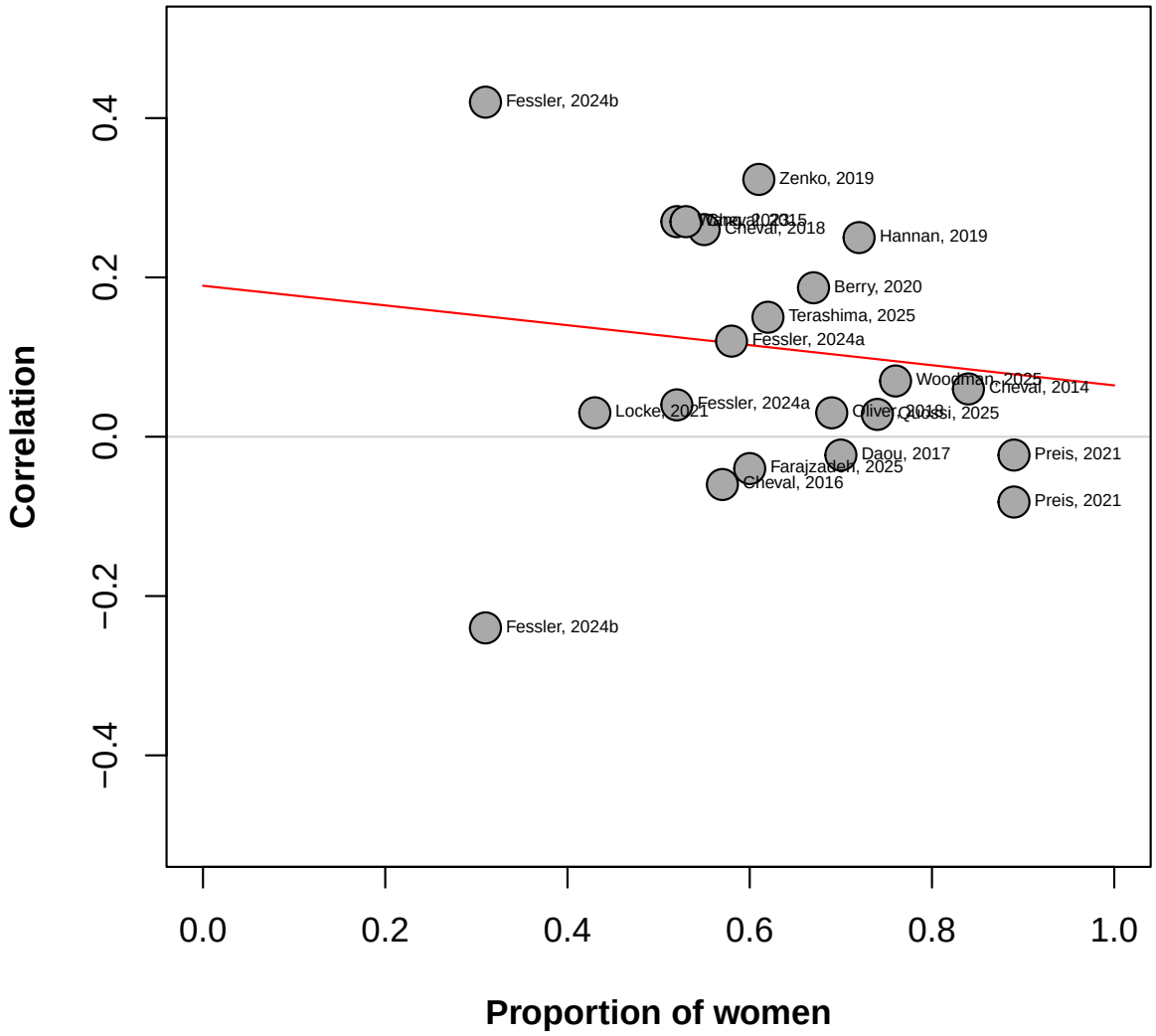
Propwomen.bubble <- bubble(m.cor.reg.women,
       xlim = c(0, 1),
       ylim = c(-0.5, 0.5),
       xlab = 'Proportion of women',
       font.lab = 2,
       studlab = TRUE,
       cex = 2,
       cex.studlab = 0.5,
       pos.studlab = 4,
       offset = 0.5,
       col.line = 'red')

Propwomen.bubble
```

```
## NULL
```

```
dev.off()
```

```
## pdf
##   2
```



Sensitivity analysis: meta-regression by quality score

```
m.cor.reg.axis <- metareg(m.cor, ~ Quality_score_10)
m.cor.reg.axis
```

```
##
## Multivariate Meta-Analysis Model (k = 22; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed    factor
```

```
## sigma^2.1  0.0111  0.1051     19     no       .id
## sigma^2.2  0.0000  0.0000     22     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 20) = 41.0175, p-val = 0.0037
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 20) = 0.3670, p-val = 0.5514
##
## Model Results:
##
##                   estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt             0.1905  0.1374   1.3862  20  0.1809  -0.0962  0.4772
## Quality_score_10   -0.0134  0.0221  -0.6058  20  0.5514  -0.0594  0.0327
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

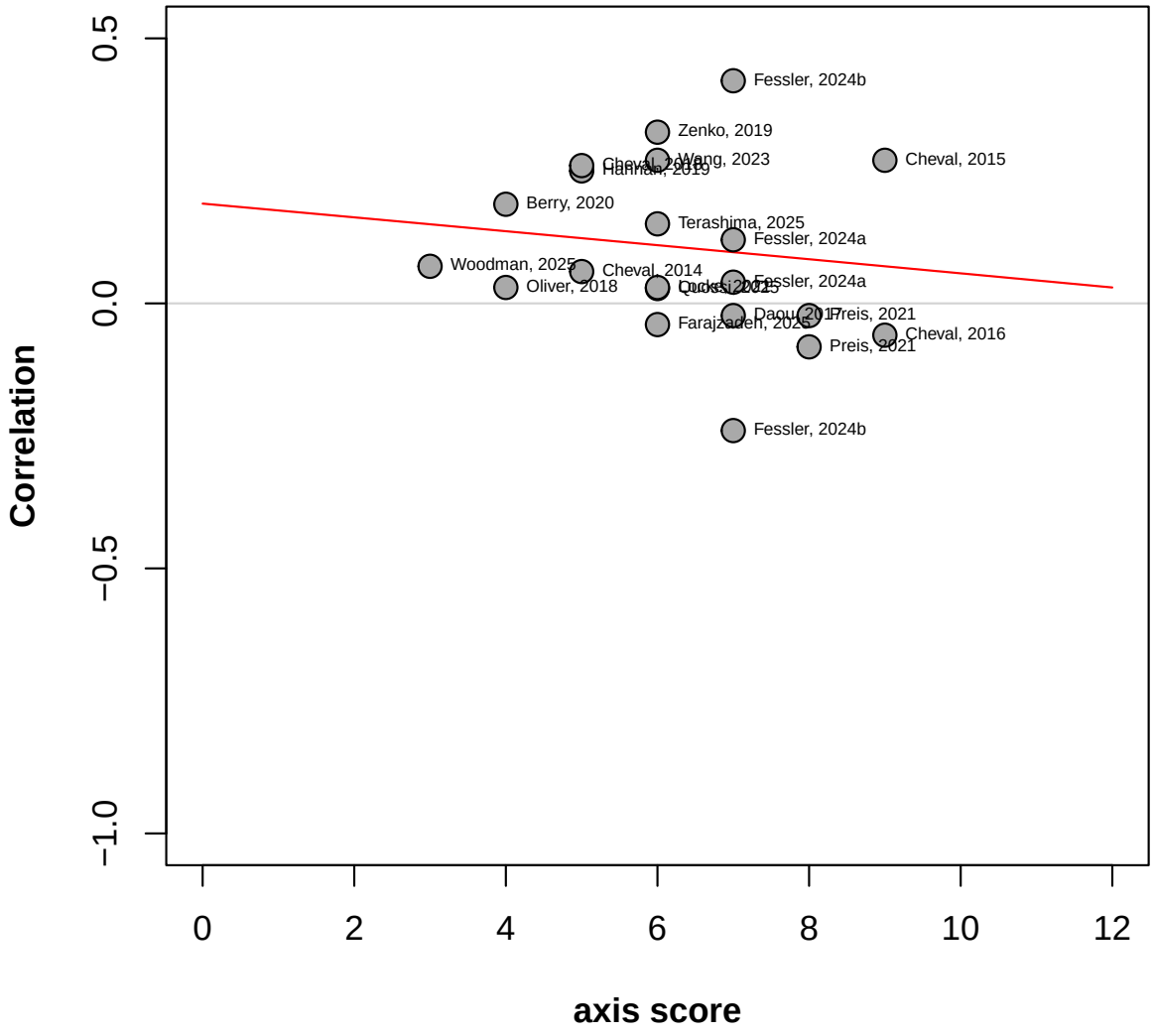
The meta-regression by quality score based on the Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies and the Quality assessment of Controlled Intervention Studies questionnaires (k = 18) showed that a study's quality did not influence correlation values (b = −0.01, 95% CI [−0.06, 0.03], t(20) = −0.61, p = 0.551).

```
pdf("figures/figure-latex/Bubble plot for meta-regression by quality score.pdf",
    width = 6, height = 6, onefile = FALSE)
Axis.bubble <- bubble(m.cor.reg.axis,
                xlim = c(0, 12),
                ylim = c(-1.0, 0.5),
                xlab = 'axis score',
                studlab = TRUE,
                font.lab = 2,
                cex = 1.5,
                cex.studlab = 0.5,
                pos.studlab = 4,
                offset = 0.5,
                col.line = 'red')
Axis.bubble
```

```
## NULL
```

```
dev.off()
```

```
## pdf
##   2
```

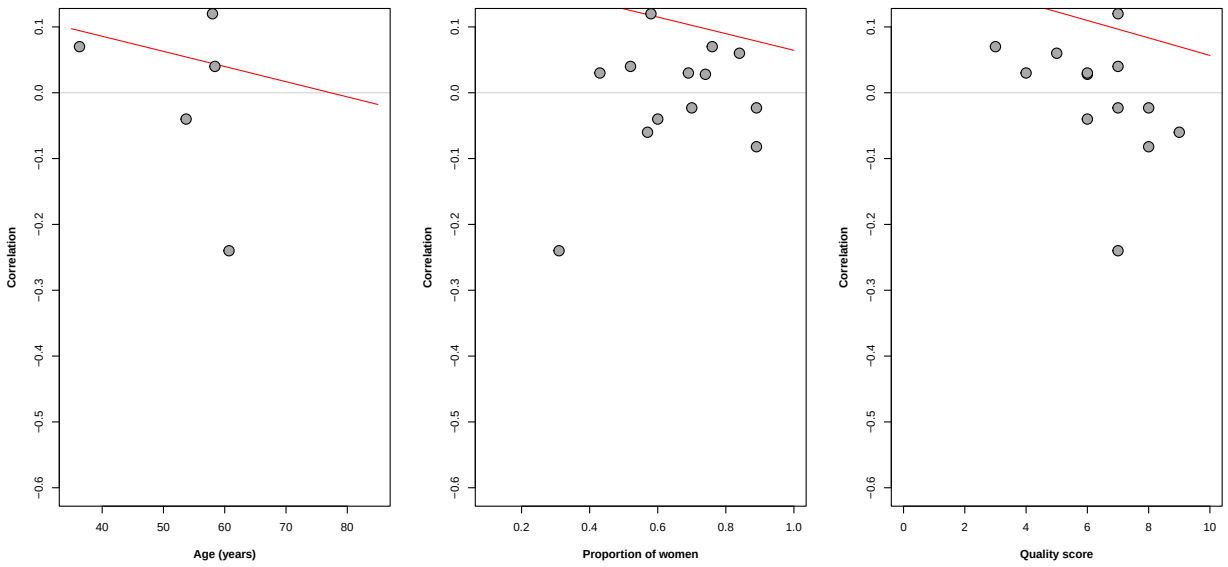



```
parameter <- par(mfrow=c(1,3))

parameter <- bubble(m.cor.reg.age,
                    xlim = c(35, 85),
                    ylim = c(-0.6, 0.1),
                    xlab = 'Age (years)',
                    font.lab = 2,
                    studlab = F,
                    cex = 2,
                    cex.studlab = 0.5,
                    pos.studlab = 4,
                    offset = 0.5,
                    col.line = 'red')
```

```r
parameter <- bubble(m.cor.reg.women,
                    xlim = c(0.1, 1),
                    ylim = c(-0.6, 0.1),
                    xlab = 'Proportion of women',
                    font.lab = 2,
                    studlab = F,
                    cex = 2,
                    cex.studlab = 0.5,
                    pos.studlab = 4,
                    offset = 0.5,
                    col.line = 'red')

parameter <- bubble(m.cor.reg.axis,
                   xlim = c(0, 10),
                   ylim = c(-0.6, 0.1),
                   xlab = 'Quality score',
                   studlab = F,
                   font.lab = 2,
                   cex = 2,
                   cex.studlab = 0.5,
                   pos.studlab = 4,
                   offset = 0.5,
                   col.line = 'red')
```



Publication bias analysis

Small-study effects

Funnel plot

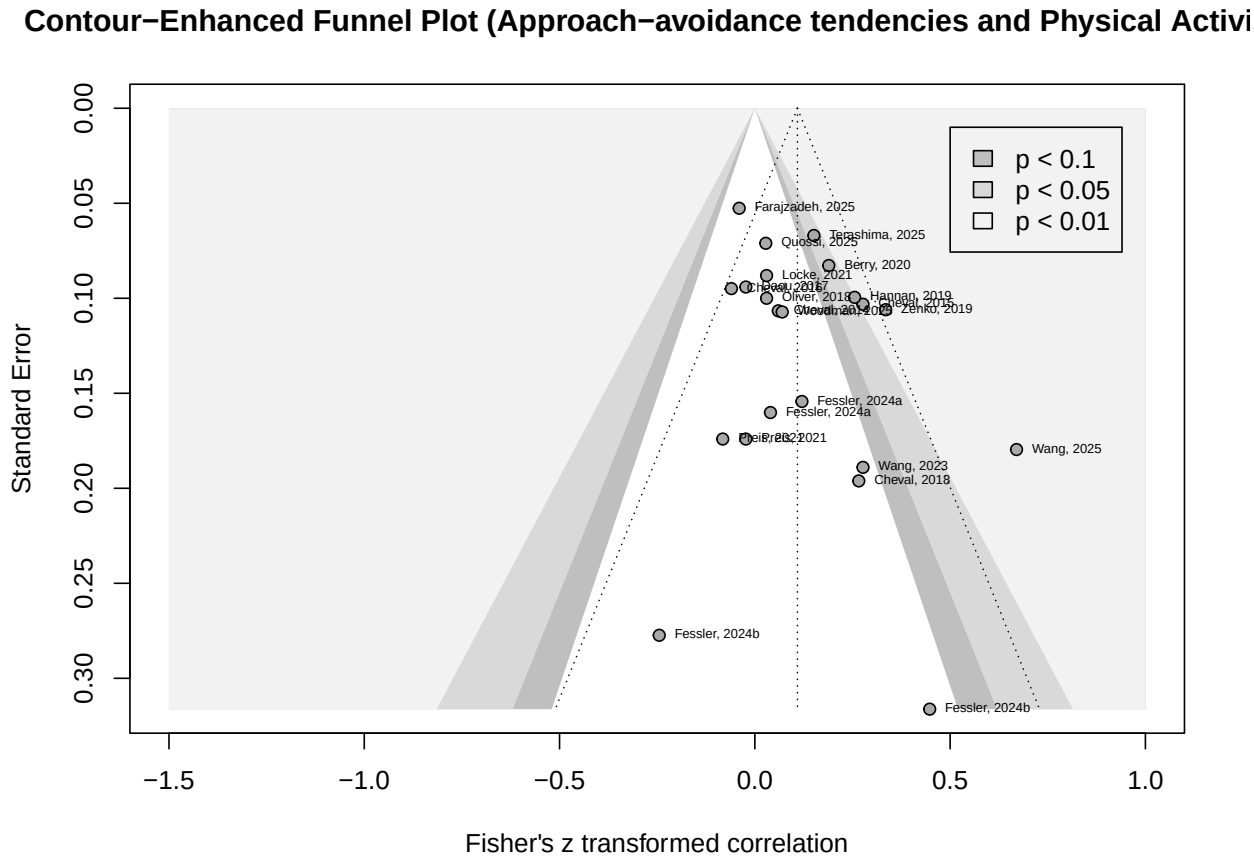
```
## Warning in plot.window(...): "offset" is not a graphical parameter
```

```
## Warning in plot.xy(xy, type, ...): "offset" is not a graphical parameter

## Warning in axis(side = side, at = at, labels = labels, ...): "offset" is not a
## graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "offset" is not a
## graphical parameter

## Warning in box(...): "offset" is not a graphical parameter

## Warning in title(...): "offset" is not a graphical parameter
```




```
col.contour = c("gray85", "gray75")

pdf("figures/figure-latex/Contour-enhanced funnel plot for main analysis no labels.pdf",
    width = 8, height = 6, onefile = FALSE)

funnel(m.cor,
       xlim = c(-1.5, 1),
       contour = c(0.95, 0.99),
       studlab = FALSE,
       cex = 1,
       cex.studlab = 0.5,
```

```
       pos.studlab = 4,
       offset = 0.5,
       col.contour = col.contour)
```

```
## Warning in plot.window(...): "offset" is not a graphical parameter

## Warning in plot.xy(xy, type, ...): "offset" is not a graphical parameter

## Warning in axis(side = side, at = at, labels = labels, ...): "offset" is not a
## graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "offset" is not a
## graphical parameter

## Warning in box(...): "offset" is not a graphical parameter

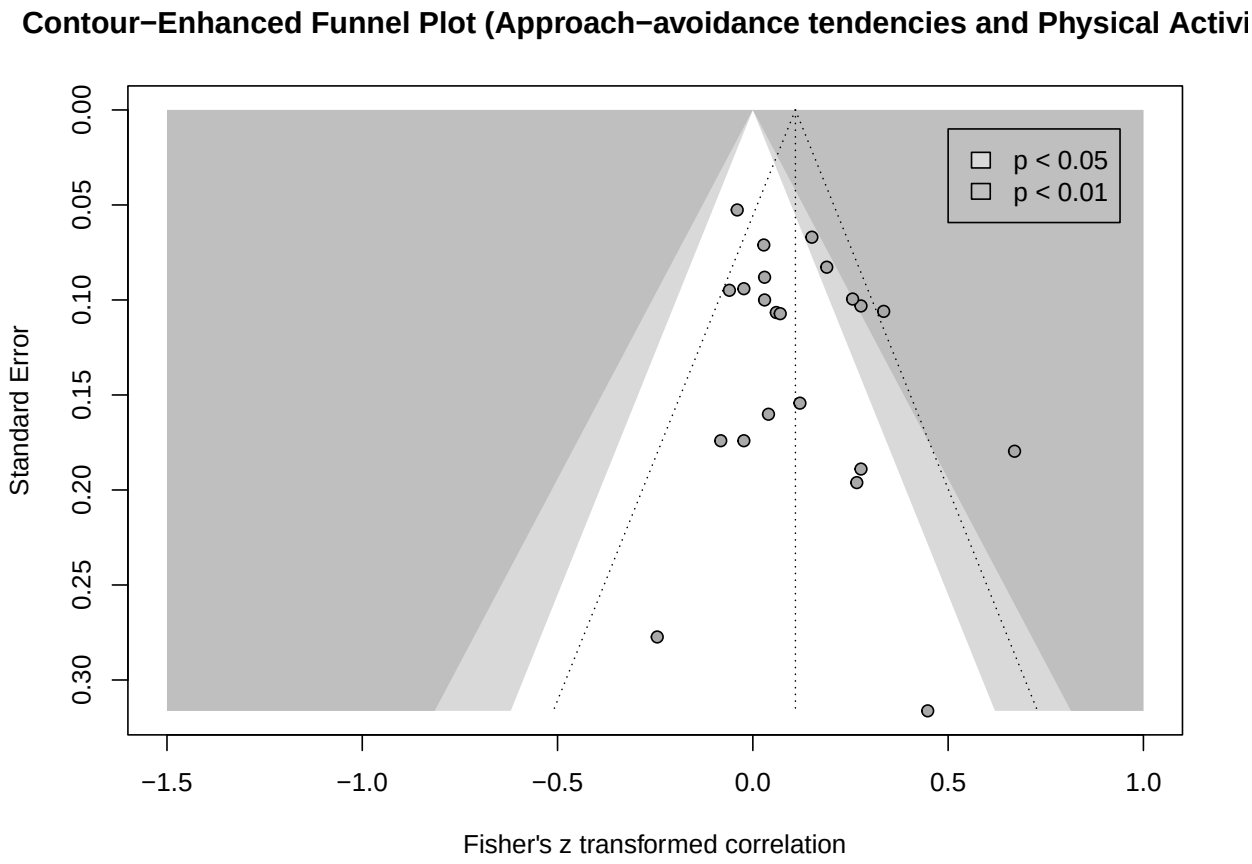
## Warning in title(...): "offset" is not a graphical parameter
```

```
legend(x =0.5, y = 0.01,
       legend = c("p < 0.05", "p < 0.01"),
       fill = col.contour)

title("Contour-Enhanced Funnel Plot (Approach-avoidance tendencies and Physical Activity)")

dev.off()
```

```
## pdf
##   2
```



Egger's test

Because the primary meta-analysis accounted for dependent effect sizes using a model with clustering, Egger's test and Rücker's limit meta-analysis could not be applied directly, as these methods are not implemented for these 3-level models. We therefore retained the model with clustering as the primary analysis and conducted a separate two-level sensitivity analysis for publication bias assessment. For the aggregated study-level sensitivity analyses, the sample size assigned to each aggregated effect corresponded to the largest sample size among effect sizes from the same study cluster, as these effects were based on overlapping samples.

```
dat_es <- escalc(measure = "GEN", yi = z, vi = var.z, data = AA_r)

dat_agg <- aggregate(
  dat_es,
  cluster = Cluster,
  struct = "CS",
  rho = 0.50,
  weighted = TRUE
)

dat_agg$Cor_agg <- tanh(dat_agg$yi)

n_by_cluster <- AA_r %>%
  group_by(Cluster) %>%
```

```
  summarise(
    n_agg = max(n, na.rm = TRUE),
    .groups = "drop"
  )

dat_agg <- dat_agg %>%
  left_join(n_by_cluster, by = "Cluster")

# 2-level meta-analysis with 1 effect per cluster
m.cor.2l <- metacor(
  cor = Cor_agg,
  n = n_agg,
  studlab = Cluster,
  data = dat_agg,
  random = TRUE,
  common = FALSE,
  method.tau = "REML",
  hakn = TRUE,
  prediction = TRUE
)
```

```
## Warning: Use argument 'method.random.ci' instead of 'hakn' (deprecated).
```

```
metabias(m.cor.2l, method.bias = "linreg", k.min = 9)
```

```
## Linear regression test of funnel plot asymmetry
##
## Test result: t = 1.88, df = 17, p-value = 0.0767
## Bias estimate: 1.6307 (SE = 0.8654)
##
## Details:
## - multiplicative residual heterogeneity variance (tau^2 = 1.8670)
## - predictor: standard error
## - weight:    inverse variance
## - reference: Egger et al. (1997), BMJ
```

```
eggers.test(m.cor.2l)
```

```
## Eggers' test of the intercept
## =============================
##
##  intercept       95% CI     t          p
##      1.631 -0.07 - 3.33 1.884 0.07674824
##
## Eggers' test does not indicate the presence of funnel plot asymmetry.
```

Egger's regression test showed that the data in the funnel plot was not asymmetric (b = 1.63, 95% CI: -0.07 to 3.33, p = 0.077).

Limit meta-analysis

```
limitmeta(m.cor.2l)
```

```
## Result of limit meta-analysis:
##
##  Random effects model    COR            95%-CI    z   pval
##     Adjusted estimate 0.0416 [-0.0760; 0.1581] 0.69 0.4882
##   Unadjusted estimate 0.1122 [ 0.0364; 0.1866] 3.11 0.0061
##
## Quantifying heterogeneity:
## tau^2 = 0.0106; I^2 = 53.1% [21.0%; 72.2%]; G^2 = 99.8%
##
## Test of heterogeneity:
##      Q d.f. p-value
##  38.37   18  0.0035
##
## Test of small-study effects:
##   Q-Q' d.f. p-value
##   6.63    1  0.0100
##
## Test of residual heterogeneity beyond small-study effects:
##     Q' d.f. p-value
##  31.74   17  0.0162
##
## Adjustment method: expectation (beta0)
```

The bias-corrected estimate of the true effect size, calculated using Rücker's limit meta-analysis method, indicated that the correlation was no longer statistically significant after adjustment for small-study effects (r = .04, 95% CI: −.08, .16, p = .488). The test for small-study effects was statistically significant (Q − Q = 6.63, p = .010), suggesting the presence of such effects. Residual heterogeneity remained significant after adjustment (Q = 31.74, p = .016), indicating that small-study effects alone do not fully account for the observed variability. These results may reflect publication bias, but could also be explained by other factors, such as differences in study procedures and between-study heterogeneity.

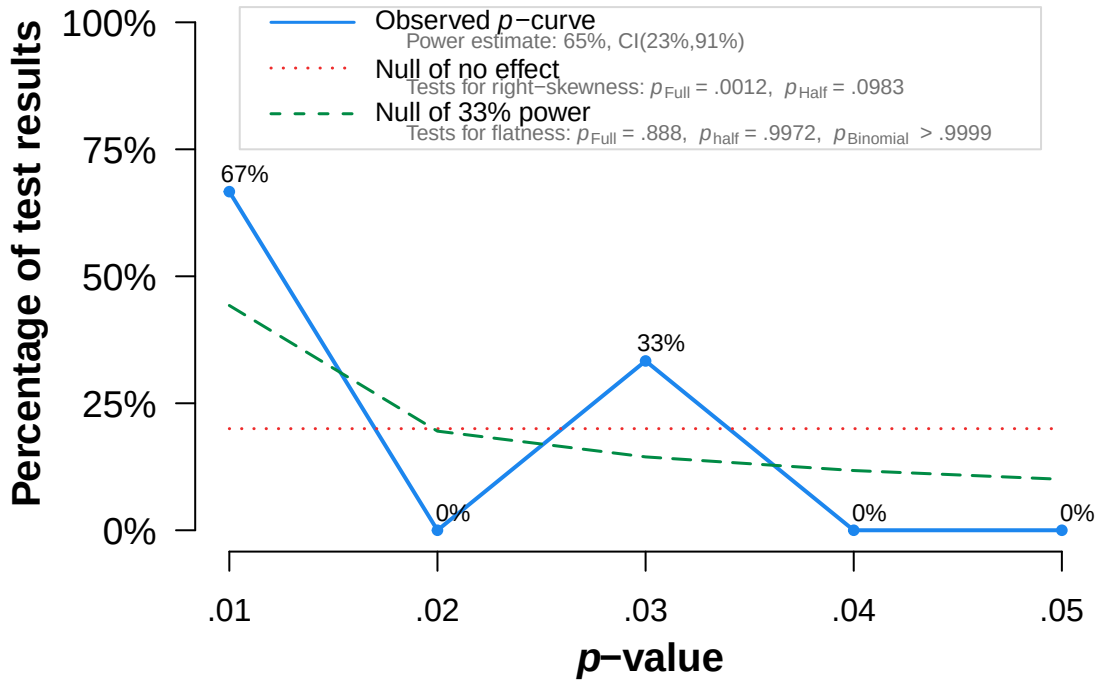
Pcurve analysis

Standard errors of the correlation values were calculated using the formula: sqrt((1 - correlation_coefficient^2) / (sample_size - 2))

```
p_curve_data = data.frame("studlab" = c(paste("Study_", 1:24, sep = "")),
                  "TE" = c(0.06, 0.27, -0.06, -0.023, 0.26, 0.03, 0.25, 0.323, 0.187, -0.023, -0.082, 0.0

                   "seTE" = c(0.106, 0.099, 0.094, 0.094, 0.186, 0.099, 0.096, 0.1, 0.081, 0.171, 0.171,

pcurve(p_curve_data)
```



Note: The observed p−curve includes 6 statistically significant (p < .05) results, of which 6 are p < .025. There were 18 additional results entered but excluded from p−curve because they were p > .05.

```
## P-curve analysis
##  ------------------------
## - Total number of provided studies: k = 24
## - Total number of p<0.05 studies included into the analysis: k = 6 (25%)
## - Total number of studies with p<0.025: k = 6 (25%)
##
## Results
##  ------------------------
##                      pBinomial  zFull pFull  zHalf pHalf
## Right-skewness test      0.016 -3.036 0.001 -1.292 0.098
## Flatness test            1.000  1.216 0.888  2.775 0.997
## Note: p-values of 0 or 1 correspond to p<0.001 and p>0.999, respectively.
## Power Estimate: 65% (22.7%-90.6%)
##
## Evidential value
##  ------------------------
## - Evidential value present: yes
## - Evidential value absent/inadequate: no
```

The 20 studies reporting a total of 24 correlation values (Pearson's r or Spearman's) were considered for the p-curve analysis. Of these, 6 studies (25.0%) reported statistically significant results (p < .05), and all 6 had p < .025, allowing their inclusion in the p-curve. The remaining results (k = 18) were excluded because their p-values exceeded .05.

The p-curve exhibited significant right-skewness, indicating evidential value in the literature. This was supported by the test for right-skewness for the full curve (p = .0012), whereas the half-curve test was

not statistically significant (p = .0983). In addition, tests for flatness were non-significant (all p .888), suggesting that the distribution of significant findings is inconsistent with low evidential value or selective reporting alone.

The estimated statistical power of the included studies was 65% (95% CI: 23% to 91%), indicating moderate evidential value. However, this estimate should be interpreted with caution given the relatively small number of significant results (k = 6) included in the analysis.

References

- Cohen J. Statistical power analysis for the behavioral sciences. 2nd ed.; 1988.

- Egger M, Davey Smith G, Schneider M, Minder C. Bias in meta-analysis detected by a simple, graphical test. BMJ. 1997;315(7109):629-634. https://doi.org/10.1136/bmj.315.7109.62

- Higgins JPT, Thompson SG. Quantifying heterogeneity in a meta-analysis. Stat Med. 2002;21(11):1539-1558. https://doi.org/10.1002/sim.1186
- Knapp G, Hartung J. Improved tests for a random effects meta-regression with a single covariate. Stat Med. 2003;22(17):2693-2710. https://doi.org/10.1002/sim.1482

- R Core Team. R: A language and environment for statistical computing [Computer software]. Version 4.3.1. Vienna, Austria: Foundation for Statistical Computing; 2023. https://www.r-project.org
- Schwarzer G. Meta: General package for meta-analysis [R package]. Version 6.5-0; 2023a. https://cran.r-project.org/web/packages/meta/meta.pdf
- Schwarzer G, Carpenter RJ, Rücker G. Metasens: Advanced statistical methods to model and adjust for bias in meta-analysis [R package]. Version 1.5-2; 2023b. https://cran.r-project.org/web/packages/metasens/metasens.pdf

- Simonsohn U, Nelson LD, Simmons JP. P-curve: A key to the file-drawer. J Exp Psychol Gen. 2014;143(2):534-547. http://doi.org/10.1037/a0033242

- Viechtbauer W. Bias and efficiency of meta-analytic variance estimators in the random-effects model. J Educ Behav Stat. 2005;30(3):261-293. https://doi.org/10.3102/10769986030003261

- Viechtbauer W. Conducting meta-analyses in R with the metafor package. J Stat Softw. 2010;36(3):1-48. https://doi.org/10.18637/jss.v036.i03

- Viechtbauer W. Metafor: Meta-analysis package for R [R package]. Version 4.2-0; 2023. https://cran.r-pro-ject.org/web/packages/metafor/metafor.pdf

- Viechtbauer, "W. (2023). Metafor: Meta-Analysis Package for R (Version 4.2-0) [R pakage]. https://cran.r-project.org/web/packages/metafor/metafor.pdf